OPEN ENDED PROJECT REPORT

On

CustomerLens: Customer Segmentation Using K-Means and K-Medoids Clustering

DIV – III

SUBJECT: Data Mining and Data warehousing

Team Members:

Mevada Drashtant Pareshkumar
(ET24BTCO176)

Nirav Panwala
(ET24BTCO181)

Priyanshu Makani
(ET24BTCO170)

Merit Patel
(ET24BTCO185)

Heet Sutariya
(ET24BTCO204)

ABSTRACT

Customer segmentation is an important application of data mining in which customers are grouped according to similarities in their characteristics and behaviour. This Open Ended Project, CustomerLens, implements an end-to-end clustering workflow on the Mall Customer Segmentation dataset of 200 customer records. The workflow covers preprocessing, K-Means clustering, K-Medoids clustering using Partitioning Around Medoids (PAM), optimal-K selection, a two-dimensional Annual Income versus Spending Score visualisation, cluster profiling and algorithm comparison.

Annual Income (k$) and Spending Score (1-100) were used as clustering features after standardisation with StandardScaler. Values of K from 2 to 10 were evaluated using the Silhouette Score and the Davies-Bouldin Index, together with the algorithm-specific objective curves. Both algorithms gave their best silhouette and Davies-Bouldin values at K = 5. At K = 5, K-Means obtained a Silhouette Score of 0.554657 and a Davies-Bouldin Index of 0.572236, while K-Medoids obtained 0.556218 and 0.566089. The Adjusted Rand Index between the two partitions was 0.9597, indicating very high agreement. The results are exposed through a Streamlit application and a React analytics dashboard, and eight automated tests validate the configuration and customer assignments.

Keywords: Customer Segmentation, Clustering, K-Means, K-Medoids, PAM, Silhouette Score, Davies-Bouldin Index, Elbow Method, Optimal K, Data Preprocessing.

1. INTRODUCTION

Customer segmentation divides a customer population into groups whose members share similar characteristics. In a retail setting it helps an organisation understand different combinations of purchasing potential and engagement. Because the groups are not supplied as class labels, clustering, an unsupervised learning technique, is the appropriate approach.

This Open Ended Project was carried out in the Department of Computer Engineering, Sarvajanik College of Engineering & Technology (SCET), Sarvajanik University, as part of the subject Data Mining and Data Warehousing (BTCO14505). CustomerLens applies K-Means and K-Medoids to a mall customer dataset, determines a suitable number of clusters, compares the two algorithms and presents the results through interactive analytical interfaces. The project does not predict future customer behaviour; it describes structure present in the existing data.

1.1 Problem Statement

To develop a customer segmentation system that preprocesses customer data, determines a suitable number of clusters, applies K-Means and K-Medoids clustering, evaluates the resulting partitions and visualises the customer groups using Annual Income and Spending Score.

1.2 Objectives

- Perform data-quality checks and preprocessing of the customer dataset.
- Explore the data using exploratory data analysis (EDA).
- Select and standardise suitable clustering features.
- Evaluate candidate cluster counts from K = 2 to K = 10 (Elbow / bend-in-the-knee analysis together with silhouette and Davies-Bouldin measures).
- Implement K-Means clustering.
- Implement K-Medoids clustering using PAM.
- Create 2D scatter plots of Annual Income versus Spending Score, colour-coded by cluster.
- Profile the clusters and compare the two algorithms.
- Provide interactive interfaces for exploring the results.

1.3 Scope

The project focuses on unsupervised segmentation using two numerical variables. CustomerID is treated as an identifier. Age and Gender are retained for post-clustering profiling and are not primary clustering dimensions.

2. DATASET AND DATA UNDERSTANDING

The project uses the Mall Customer Segmentation dataset (file Mall_Customers.csv) containing 200 customer records and five attributes. The dataset includes both Annual Income and Spending Score, which are required for the primary two-dimensional visualisation. Of the 200 customers, 112 are female and 88 are male; ages range from 18 to 70 (mean 38.85), annual income ranges from 15 to 137 k$ (mean 60.56) and the spending score ranges from 1 to 99 (mean 50.20).

Important note: the Spending Score is a behavioural score supplied by the dataset. It is not a direct measurement of customer expenditure.

Table 1: Dataset schema and role of each attribute

Attribute	Type	Role in project
CustomerID	Integer	Identifier; excluded from clustering
Gender	Categorical	Retained for profiling
Age	Integer	Retained for profiling
Annual Income (k$)	Numeric	Primary clustering feature
Spending Score (1-100)	Numeric	Primary clustering feature

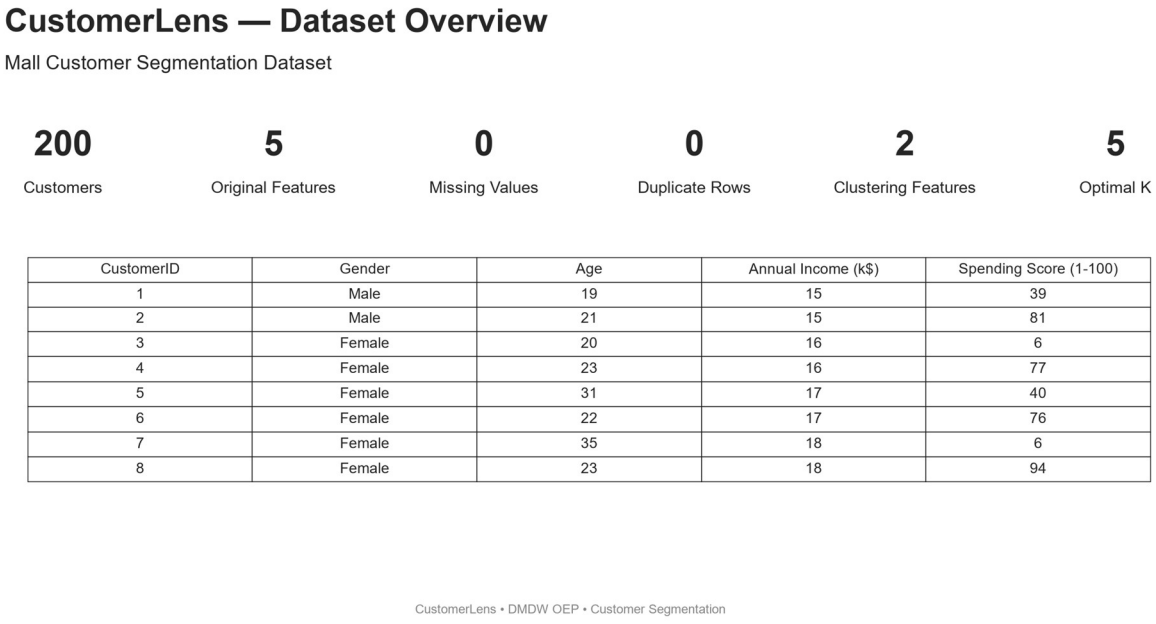


Figure 1: Dataset overview generated by the project (200 customers, 5 original features, 0 missing values, 0 duplicate rows)

3. DATA PREPROCESSING

Distance-based clustering requires clean, comparable numerical inputs. The preprocessing steps implemented in the project are summarised in Table 2.

Table 2: Preprocessing summary

Step	Outcome
Missing-value validation	No missing values in any column
Duplicate validation	No duplicate customer IDs or duplicate rows
Feature selection	Annual Income (k$) and Spending Score (1-100)
Identifier handling	CustomerID excluded from clustering
Profiling attributes	Gender and Age retained for post-clustering analysis
Scaling	StandardScaler (zero mean, unit variance); fitted scaler saved
Train/test split	Not used: the task is unsupervised

Annual Income and Spending Score are measured on different numerical ranges. StandardScaler is applied so that both features contribute comparably to Euclidean distance calculations. The same standardised matrix is used for both algorithms. The processed datasets (customerlens_final.csv, customers_audited.csv, customers_with_cluster_assignments.csv, final_customer_segmentation.csv and clustering_features_scaled.csv) make the transformations inspectable.

4. EXPLORATORY DATA ANALYSIS

Figure 2 shows the distributions of Age, Annual Income and Spending Score. Age is concentrated between the early twenties and fifties; income is right-skewed with a few high values; the spending score spans the full 1 to 99 range with a peak near the middle.

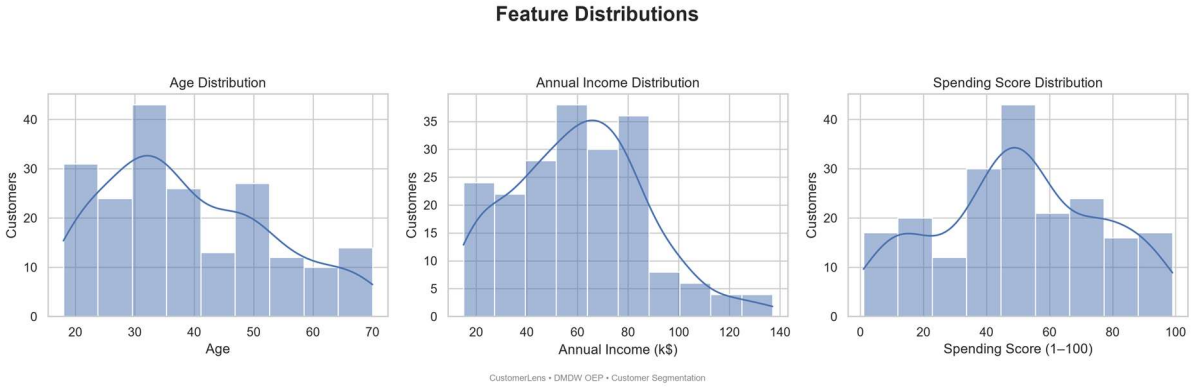


Figure 2: Distributions of Age, Annual Income and Spending Score

Figure 3 shows the raw Annual Income versus Spending Score scatter and the correlation heatmap. The correlation between income and spending score is approximately 0.01, and between age and spending score approximately -0.33. The absence of a linear relationship between the two clustering features, together with visible groupings in the scatter plot, motivates the use of clustering.

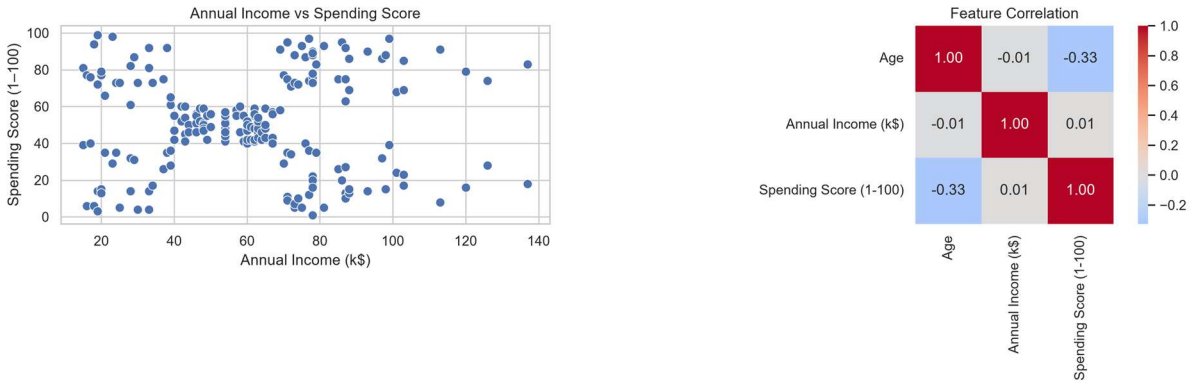


Figure 3: Exploratory relationships: Income vs Spending Score scatter and feature correlation

5. METHODOLOGY

5.1 Workflow

The project follows a sequential data-mining workflow: dataset, data-quality validation, feature selection, StandardScaler, optimal-K evaluation, K-Means and K-Medoids clustering, cluster validation, cluster profiling, business interpretation, and the CustomerLens dashboards. Optimal-K evaluation and final clustering are performed on the same standardised feature matrix.

5.2 K-Means

K-Means partitions the data into K clusters by repeatedly assigning each observation to the nearest centroid and updating each centroid as the mean of its members, minimising the within-cluster sum of squared distances (inertia). The project uses scikit-learn KMeans with k-means++ initialisation, n_init = 20 and random_state = 42.

5.3 K-Medoids using PAM

K-Medoids is similar in spirit but uses actual observations, called medoids, as cluster representatives, and minimises the sum of distances from each point to its nearest medoid. The project implements Partitioning Around Medoids (PAM) in NumPy on a pairwise Euclidean distance matrix of the standardised features. The BUILD phase selects the first medoid as the point with minimum total distance to all others and adds the remaining medoids greedily; the SWAP phase then exchanges medoids with non-medoids while the total cost decreases (maximum 100 iterations).

K-Means inertia is a sum of squared distances, whereas the K-Medoids cost is a sum of (non-squared) distances to medoids. They are different objective functions and are therefore not compared with each other in this report.

5.4 Evaluation Measures

- Silhouette Score: compares how close each point is to its own cluster with its closeness to the nearest other cluster; higher values indicate better-defined clusters.
- Davies-Bouldin Index: average similarity between each cluster and its most similar cluster; lower values indicate better separation.
- Adjusted Rand Index (ARI): chance-adjusted agreement between two partitions. It compares the K-Means and K-Medoids results with each other and is not classification accuracy.

6. OPTIMAL K SELECTION

K values from 2 to 10 were evaluated for both algorithms. For each K, the Silhouette Score and Davies-Bouldin Index were recorded together with the algorithm-specific objective (K-Means inertia and K-Medoids cost). The results are listed in Table 3 and plotted in Figure 4.

Table 3: Optimal-K evaluation (K = 2 to 10)

K	Silh. K-Means	Silh. K-Medoids	DB K-Means	DB K-Medoids	Inertia	Medoid cost
2	0.3213	0.3916	1.2670	0.9540	269.69	191.62
3	0.4666	0.4641	0.7165	0.7230	157.70	147.04
4	0.4939	0.4857	0.7096	0.7218	108.92	123.15
5	0.5547	0.5562	0.5722	0.5661	65.57	100.72
6	0.5399	0.4558	0.6546	0.6700	55.06	89.85
7	0.5281	0.4516	0.7148	0.7341	44.86	82.99
8	0.4567	0.4544	0.7579	0.7349	37.15	77.77
9	0.4571	0.4252	0.7632	0.7736	32.39	73.67
10	0.4362	0.4347	0.7645	0.7909	29.69	67.49

Silh. = Silhouette (higher is better); DB = Davies-Bouldin (lower is better). Inertia and medoid cost are different objectives. K = 5 gives the highest Silhouette and lowest Davies-Bouldin for both algorithms.

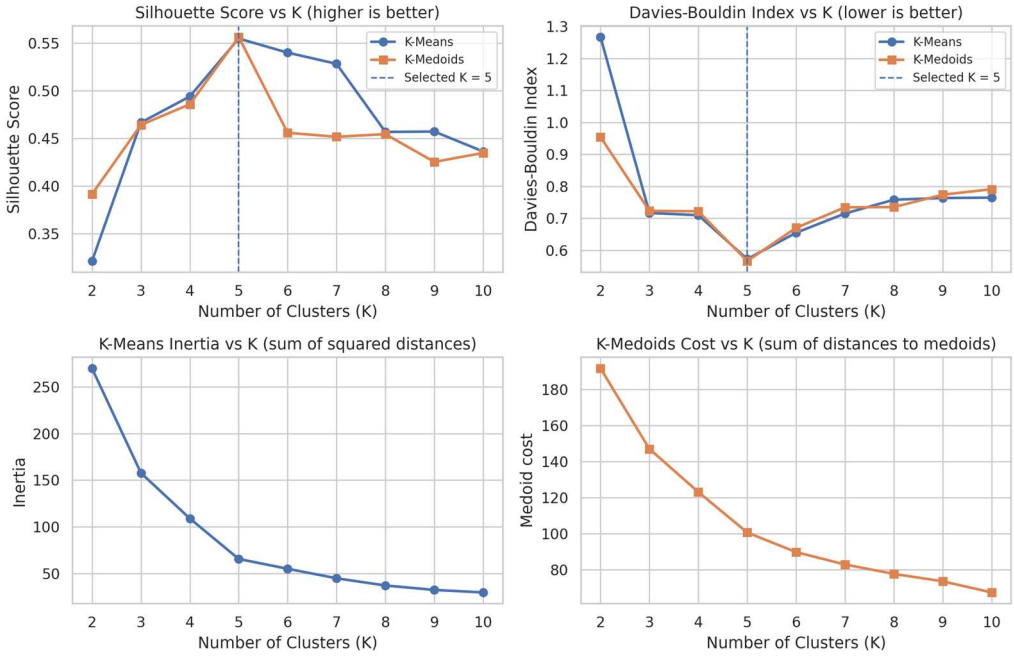


Figure 4: Silhouette Score, Davies-Bouldin Index, K-Means inertia and K-Medoids cost versus K (plotted from the project result CSVs; inertia and medoid cost are separate objectives)

Both algorithms reach their highest Silhouette Score at K = 5 (0.5547 for K-Means, 0.5562 for K-Medoids) and their lowest Davies-Bouldin Index at K = 5 (0.5722 and 0.5661). The objective curves decrease steadily as K increases and show a bend near K = 5, but the elbow alone is not decisive; the selection of K = 5 rests on the agreement of the silhouette and Davies-Bouldin measures with the elbow-style curve.

7. EXPERIMENTAL RESULTS

7.1 K-Means Segmentation

Figure 5 shows the required visualisation: Annual Income versus Spending Score with customers colour-coded by K-Means cluster and the centroids marked with black crosses. Five well-separated groups are visible.

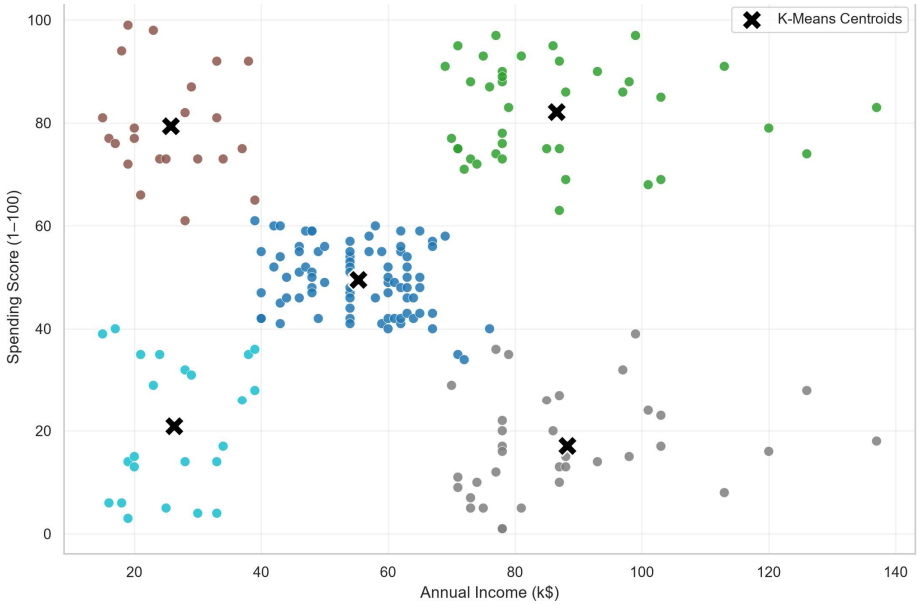


Figure 5: K-Means customer segmentation (K = 5): Annual Income vs Spending Score

7.2 K-Medoids Segmentation

Figure 6 shows the K-Medoids (PAM) result from the project notebook. The five medoids are actual customers (CustomerID 81, 162, 177, 16 and 25). The cluster sizes are 84, 39, 35, 21 and 21. The K-Medoids cluster numbering is independent of the K-Means numbering.

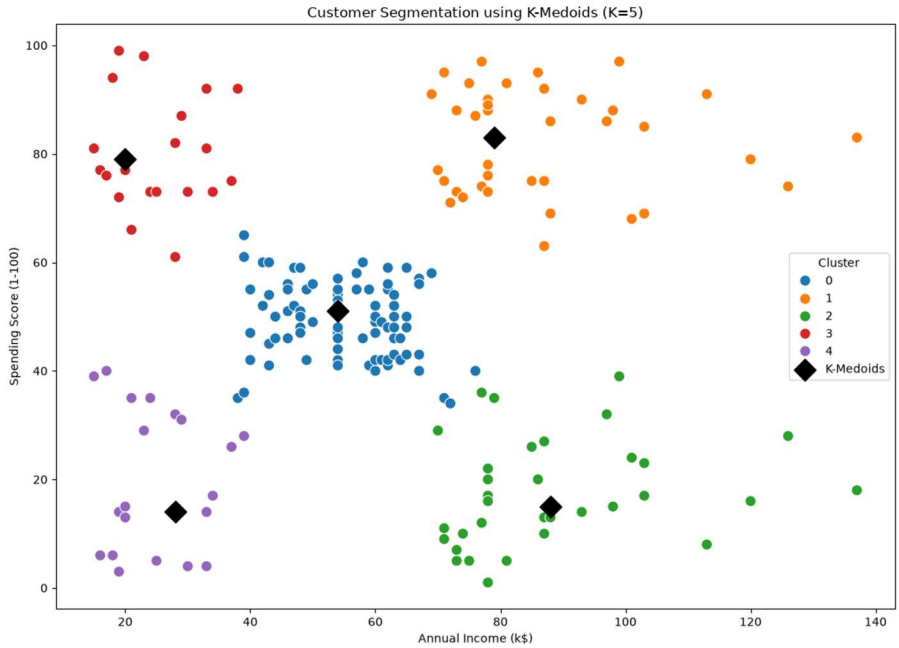


Figure 6: K-Medoids (PAM) customer segmentation (K = 5); diamonds mark the medoids

7.3 Cluster Profiles

Table 4 gives the K-Means cluster profiles. Income is in thousands of dollars and spending is the dataset score from 1 to 100.

Table 4: K-Means cluster profiles (K = 5)

Cluster	Customers	Share (%)	Avg. age	Avg. income (k$)	Avg. spending score
0	81	40.5	42.72	55.30	49.52
1	39	19.5	32.69	86.54	82.13
2	22	11.0	25.27	25.73	79.36
3	35	17.5	41.11	88.20	17.11
4	23	11.5	45.22	26.30	20.91



Figure 7: Average income and spending score by K-Means cluster

7.4 Algorithm Comparison

Table 5: Final comparison of K-Means and K-Medoids at K = 5

Metric	K-Means	K-Medoids (PAM)
K	5	5
Silhouette Score (higher is better)	0.554657	0.556218
Davies-Bouldin Index (lower is better)	0.572236	0.566089
Runtime (recorded)*	0.164474 s	0.076011 s
Adjusted Rand Index (agreement between the two partitions; a single pairwise value)	0.9597	0.9597

**Runtime is specific to this implementation, experiment and machine; K-Means was timed with 20 restarts (n_init = 20), and the K-Medoids (PAM) timing covers the PAM routine only, excluding the earlier pairwise distance-matrix computation. It does not establish that either algorithm is generally faster.*

In this experiment K-Medoids achieved a slightly higher Silhouette Score and a slightly lower Davies-Bouldin Index than K-Means. The differences are small and apply to this dataset and set-up only; they do not show that K-Medoids is universally better. The ARI of 0.9597 indicates that the two partitions are almost identical. Table 6 shows that 197 of the 200 customers fall in matching groups; three low-income customers (IDs 41, 43 and 46) assigned to K-Means clusters 4 and 2 are placed in the large central cluster by K-Medoids.

Table 6: Contingency table of cluster assignments (rows: K-Means, columns: K-Medoids)

K-Means \ K-Medoids	0	1	2	3	4
0	81	0	0	0	0
1	0	39	0	0	0
2	1	0	0	21	0
3	0	0	35	0	0
4	2	0	0	0	21

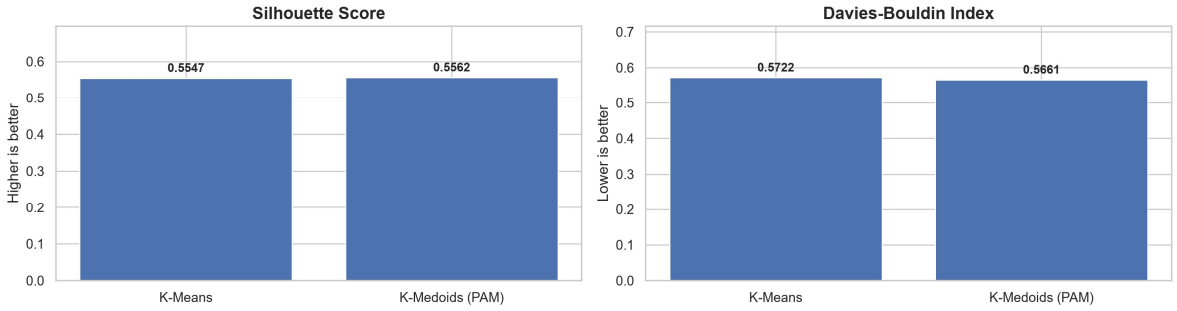


Figure 8: Silhouette Score and Davies-Bouldin Index of K-Means and K-Medoids at K = 5 (project figure; runtime panel omitted because it shows only one algorithm)

8. BUSINESS INTERPRETATION

Segment names below are descriptive analytical labels derived from each cluster's average income and spending score. They are not ground-truth customer categories, and the suggested actions are hypotheses, not tested outcomes. Because the Spending Score is a dataset-supplied behavioural score, "high spending" means a high score rather than measured expenditure.

Table 7: Descriptive segments and possible actions (K-Means)

Cluster	Segment label	Interpretation	Possible action
1	High-Value Customers	Relatively high income and high spending score	Loyalty programmes, premium offers, retention
0	Balanced / Moderate Customers	Moderate income and moderate spending	Maintain engagement; personalised promotions
2	High-Spending Budget Customers	Lower income but relatively high spending score	Affordable, value-oriented campaigns
3	High-Income Low-Spending Customers	High income but relatively low spending score	Investigate barriers to spending; targeted engagement
4	Low-Engagement Customers	Relatively low income and low spending score	Introductory offers and engagement campaigns

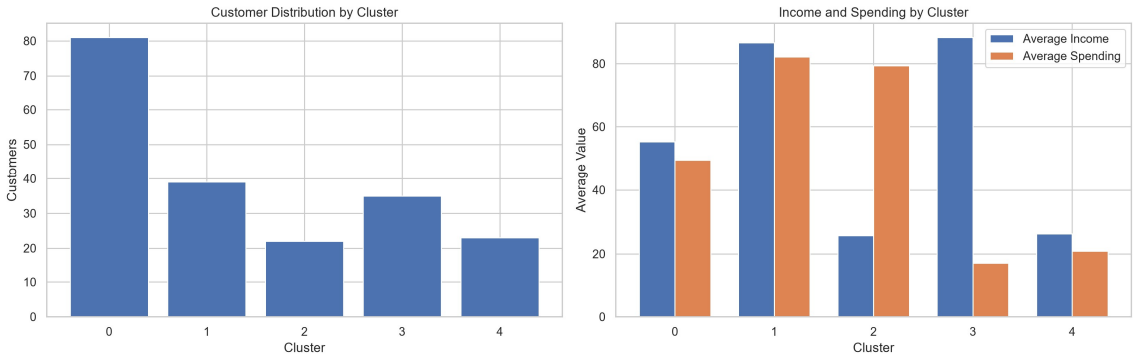


Figure 9: Customer distribution and income/spending profile by cluster

9. APPLICATION / DASHBOARD

CustomerLens provides two interfaces on top of the validated results. The Streamlit application (app.py) offers seven sections: Overview, Data Quality and EDA, Optimal K, Cluster Explorer, Algorithm Comparison, Business Insights and Customer Explorer; the last also assigns a hypothetical customer to the nearest existing cluster using the saved scaler and models. The React dashboard (React, Vite, TypeScript, Tailwind CSS and Recharts) provides the same seven pages and reads customerlens.json, which export_frontend_data.py generates from the validated Python results. The dashboard displays results and does not retrain models.

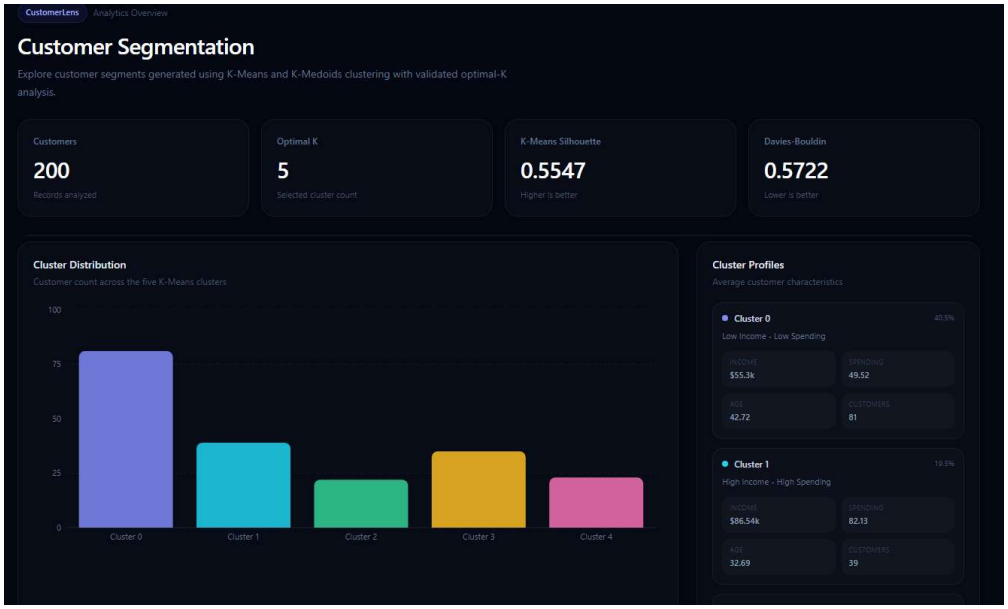


Figure 10: Overview page of the React dashboard. Note: this screenshot shows the legacy project label "Low Income - Low Spending" for Cluster 0; this report uses "Balanced / Moderate" (see Section 11).

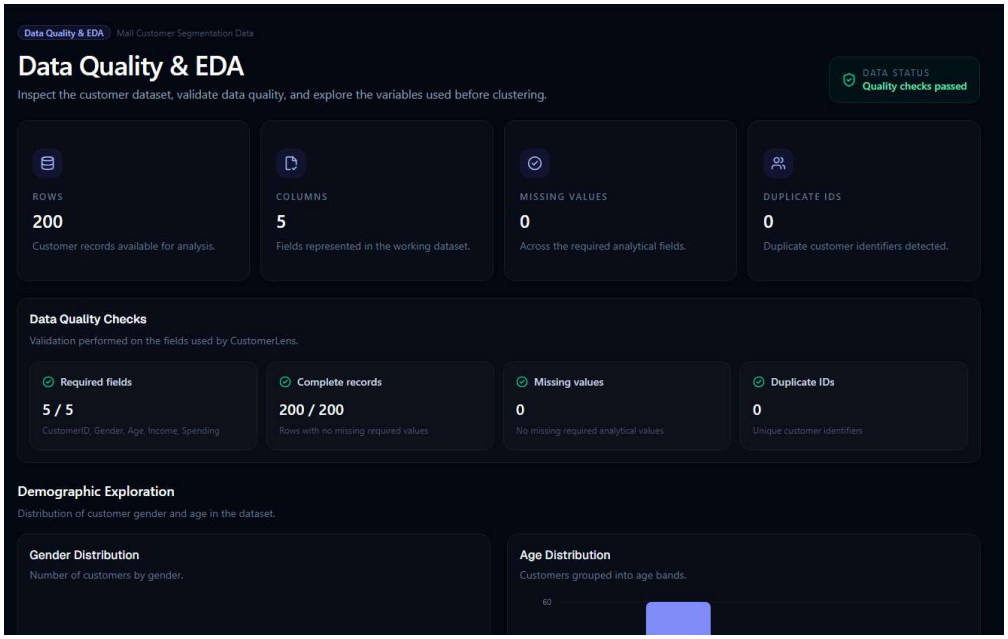


Figure 11: Data Quality page

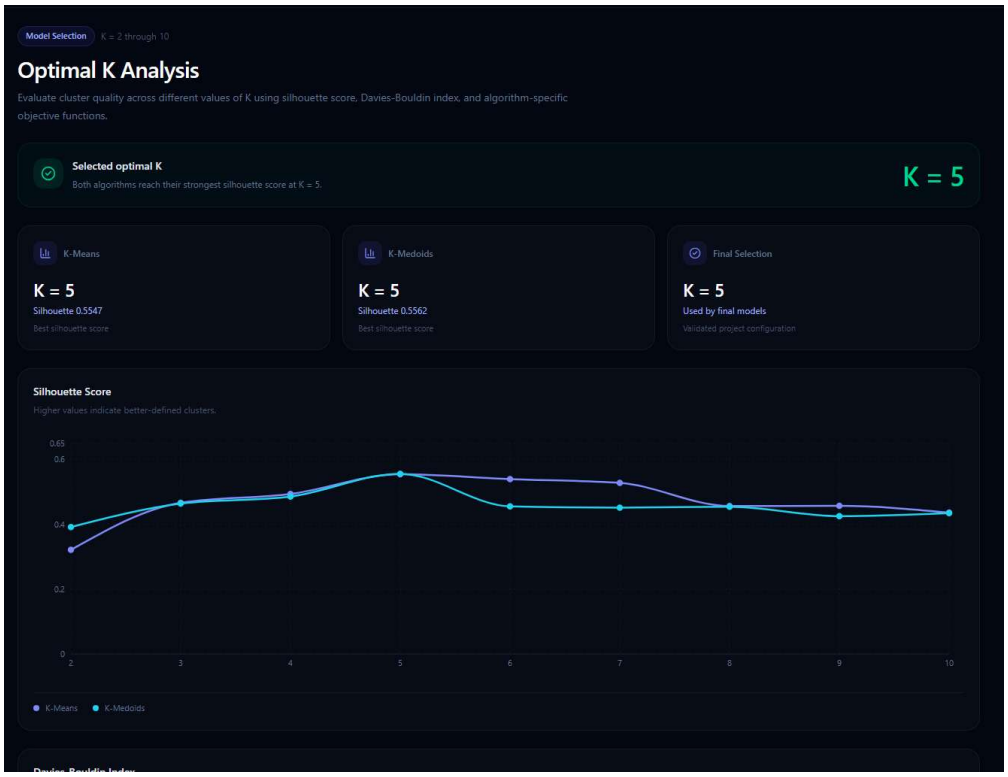


Figure 12: Optimal K page

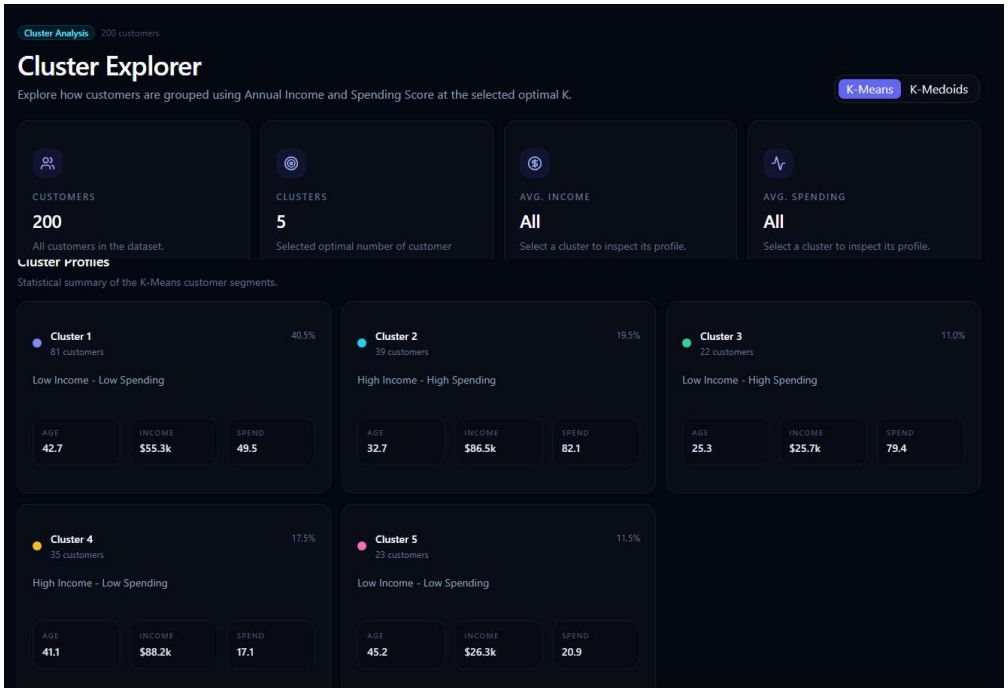


Figure 13: Cluster Explorer page. Note: this screenshot shows the legacy project label "Low Income - Low Spending" for Cluster 0; this report uses "Balanced / Moderate" (see Section 11).

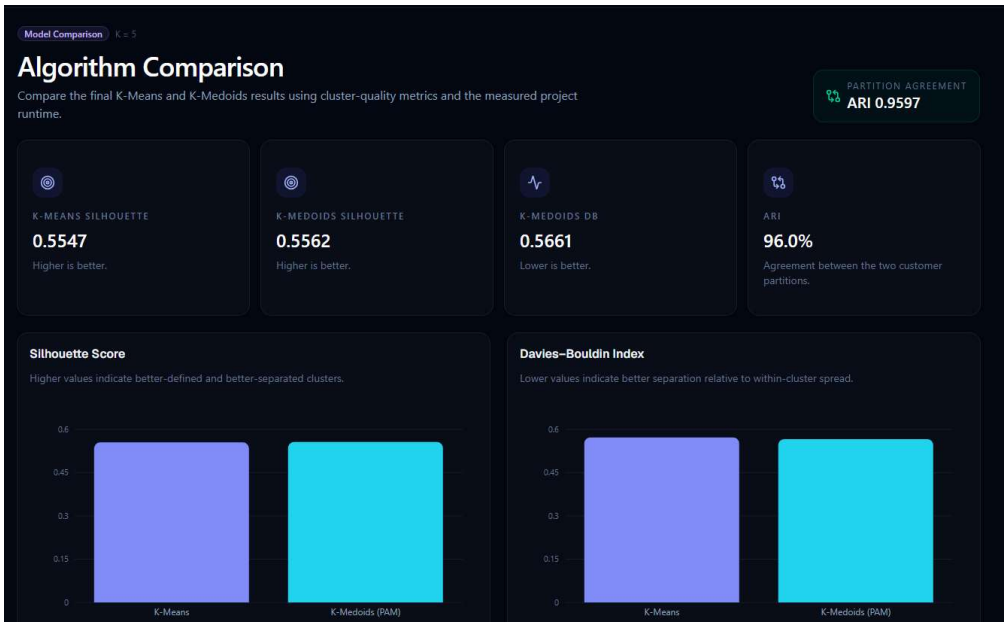


Figure 14: Algorithm Comparison page

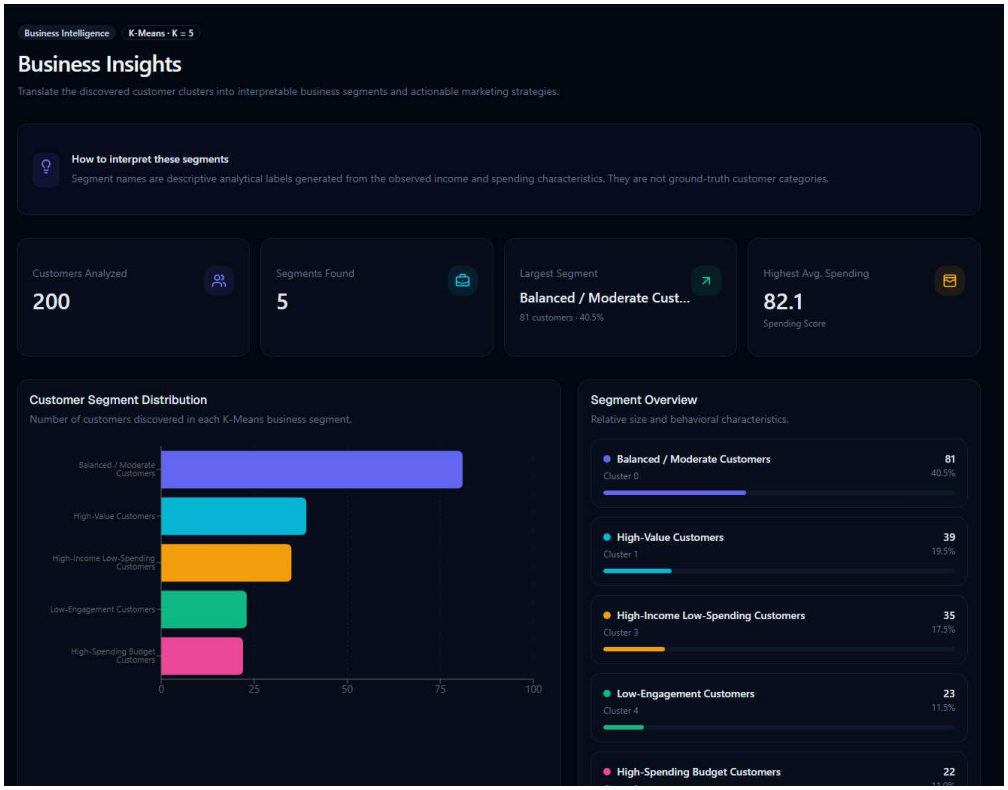


Figure 15: Business Insights page

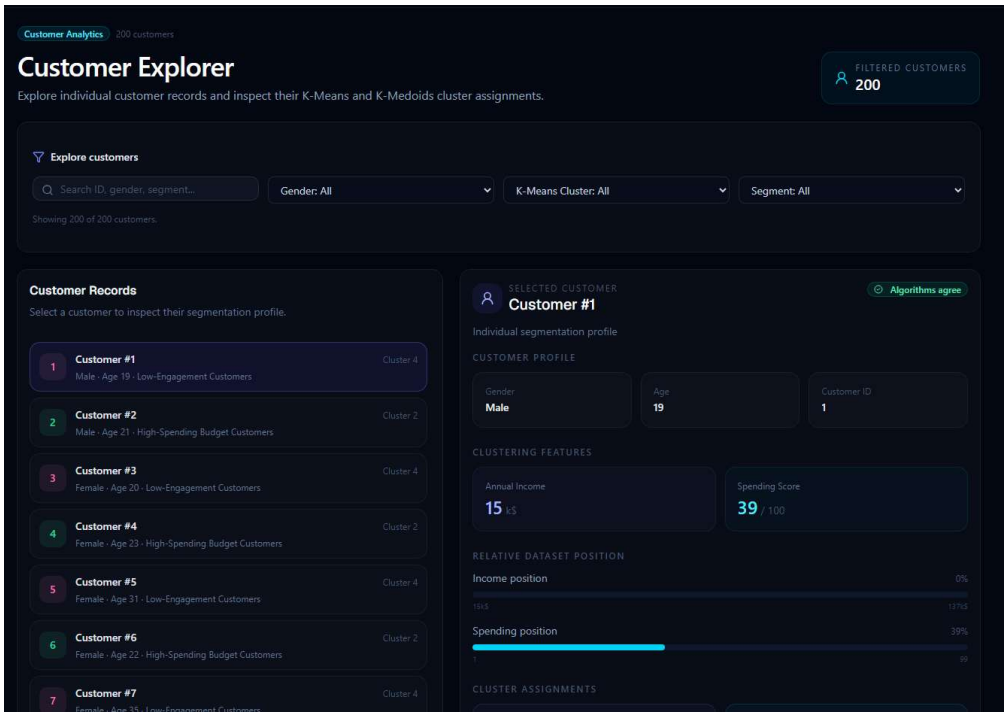


Figure 16: Customer Explorer page



Figure 17: Customer Explorer figure: an example customer shown in the cluster map (project figure)

10. TESTING AND VALIDATION

The project includes eight automated pytest tests (tests/test_clustering.py). Recorded verification status: 8 tests passed, and the frontend production build and lint passed. During the final audit, the frontend build and the eight tests (run against regenerated model files, because the trained model files were not part of the supplied package) were re-run successfully, while the lint check could not be re-run because the supplied package contains no ESLint configuration file. The tests validate configuration and assignment completeness; clustering quality is assessed separately using the Silhouette Score, Davies-Bouldin Index and ARI.

Table 8: Automated validation tests

No.	Test	Result
1	Dataset has 200 records and at least 5 columns	Passed
2	Clustering features are Annual Income and Spending Score	Passed
3	Clustering features have no missing values	Passed
4	Selected optimal K equals 5	Passed
5	K-Means model has 5 clusters	Passed
6	K-Medoids artifact has 5 medoids	Passed
7	K-Means assigns every customer to a cluster 0 to 4	Passed
8	K-Medoids assigns every customer to a cluster 0 to 4	Passed

11. LIMITATIONS

- The dataset contains only 200 customers and the primary analysis uses only two features.
- Spending Score is a behavioural score supplied by the dataset, not direct monetary spending.
- Segment names are interpretations, not predefined or ground-truth categories.
- Runtime values are specific to the implementation and machine and do not support general speed claims.
- Results may differ on other datasets or customer populations.
- The project describes existing customers and does not predict future behaviour.
- Naming inconsistency: some project material (kmeans_cluster_profiles.csv and the React Overview and Cluster Explorer pages, and the Streamlit Business Insights rule, which names it "Low-Income / Low-Spending Customers") labels Cluster 0 as "Low Income - Low Spending", although its average income (55.30 k$) and spending score (49.52) are close to the dataset means (60.56 k$ and 50.20). This report consistently uses "Balanced / Moderate" and treats all segment names as descriptive interpretations.
- K = 5 is the best-scoring value among the evaluated K = 2 to 10 for this dataset and experiment; it is not presented as a universal truth.

12. CONCLUSION

CustomerLens applies K-Means and K-Medoids (PAM) to segment 200 mall customers using Annual Income and Spending Score. Silhouette and Davies-Bouldin analysis over K = 2 to 10 selected K = 5 for both algorithms. At K = 5, K-Medoids obtained a marginally higher Silhouette Score (0.556218 versus 0.554657) and marginally lower Davies-Bouldin Index (0.566089 versus 0.572236), and the partitions agree strongly (ARI 0.9597). Five descriptive segments were profiled and presented through Streamlit and React interfaces, supported by eight passing automated tests.

13. FUTURE SCOPE

The following are proposals only and are not implemented in this project: application to larger, real-world customer data; inclusion of additional behavioural features; analysis of temporal customer behaviour; deployment and monitoring of the analytics application; and richer segmentation features.

REFERENCES

[1] MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. Proc. 5th Berkeley Symposium on Mathematical Statistics and Probability.

[2] Arthur, D. and Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. Proc. ACM-SIAM SODA.

[3] Kaufman, L. and Rousseeuw, P. J. (1990). Finding Groups in Data: An Introduction to Cluster Analysis. Wiley.

[4] Rousseeuw, P. J. (1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53-65.

[5] Davies, D. L. and Bouldin, D. W. (1979). A cluster separation measure. IEEE Transactions on Pattern Analysis and Machine Intelligence, 1(2), 224-227.

[6] Hubert, L. and Arabie, P. (1985). Comparing partitions. Journal of Classification, 2, 193-218.

[7] Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

[8] Mall Customer Segmentation Data, file Mall_Customers.csv (200 records, 5 attributes), as supplied with the CustomerLens project in data/raw/. A public listing of this dataset: Customer Segmentation Tutorial in Python (V. Choudhary), Kaggle, https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python. The exact download source and licence of the project copy are not recorded in the project files.